Evaluate $a^2 + b^2$ when $a = 6$ and $b = 8$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

Evaluate $a^2 + b^2$ when $a = 5$ and $b = 12$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

Evaluate $c^2 - a^2$ when $c = 10$ and $a = 6$.

A. 16

B. 136

C. 64

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

True or false?

When $a = 3$ and $b = 4$, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

True or false?

When $a = 2$ and $b = 3$, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 8.G.B.6

FORWARD 1

Prove the converse of the Pythagorean theorem with congruent triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has sides BC = a, AC = b, and AB = c, with $a^2 + b^2 = c^2$.

Prove that angle C is a right angle. (Build a right triangle with legs a and b.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

In that proof, which congruence criterion is used, and why does it apply?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

In that proof, which theorem gives the length of the hypotenuse of the triangle that was built?

A. The triangle angle sum theorem

B. The converse of the Pythagorean theorem

C. The Pythagorean theorem

D. The exterior angle theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

True or false?

If $a^2 + b^2 > c^2$ for the longest side c of a triangle, the angle opposite c is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

True or false?

Measuring angle C with a protractor once is a proof that a triangle with $a^2 + b^2 = c^2$ has a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 8.G.B.6

FORWARD 2

Use coordinates to show that a triangle is a right triangle • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Is the triangle with vertices (1, 1), (4, 1), and (4, 5) a right triangle? Use side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

Use slopes to decide whether the triangle with vertices (0, 0), (2, 4), and (6, 2) has a right angle at (2, 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

Which triangle is a right triangle?

A. (0, 0), (2, 2), (4, 1)

B. (0, 0), (3, 1), (1, 3)

C. (1, 1), (2, 3), (3, 1)

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

True or false?

The triangle with vertices (0, 0), (5, 0), and (0, 5) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

True or false?

The triangle with vertices (0, 0), (4, 0), and (2, 1) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in two dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Find the squares of decimals and whole numbers	**6.EE.A.1**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve right triangles with the Pythagorean theorem and trigonom…	**HSG-SRT.C.8**
FORWARD 2	Use right-triangle reasoning to make design decisions under const…	**HSG-MG.A.3**

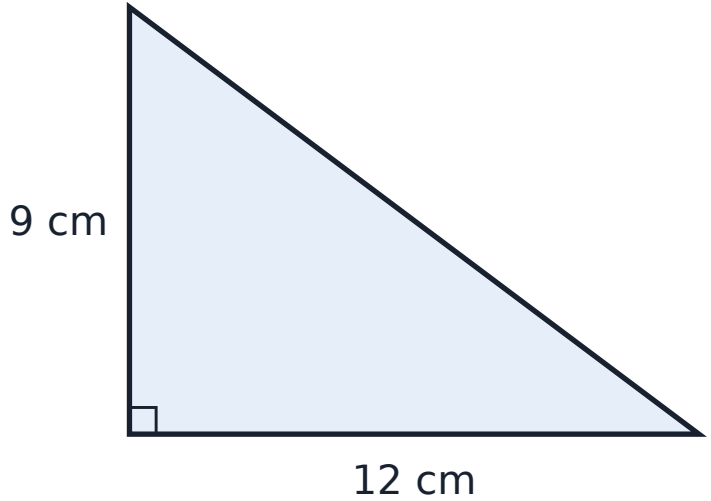
SET 46 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

Find the length of the hypotenuse.

Hypotenuse: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

A 13-foot ladder leans against a wall. Its base is 5 feet from the wall.

How high up the wall does it reach?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

A rectangular screen is 48 inches wide and 36 inches tall.

How long is its diagonal?

A. 60 inches

B. 84 inches

C. 72 inches

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

A right triangle has a hypotenuse of 17 and a leg of 8.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

True or false?

A square with 10 cm sides has a diagonal of about 14.1 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

SET 46 • 8.G.B.7

BACKWARD 1

Find the squares of decimals and whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

14^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

Find the value.

2.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

Find the value.

17^2

A. 189

B. 279

C. 289

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

$1.5^2 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 8.G.B.7

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$x + 64 = 289$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

Solve for x.

$25 + x = 169$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

Solve for x.

$81 + x = 100$

A. $x = 19$

B. $x = 181$

C. $x = 9$

D. $x = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

True or false?

$x = 15$ is the solution of $x + 49 = 64$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

True or false?

$x = 46$ is the solution of $64 + x = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 8.G.B.7

FORWARD 1

Solve right triangles with the Pythagorean theorem and trigonometric ratios • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has a hypotenuse of 10 and a 30° angle. $\sin 30° = 0.5$.

How long is the side opposite the 30° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

A ramp makes a 5° angle with the ground and rises 2 feet. Use $\sin 5° \approx 0.087$.

How long is the ramp, to the nearest foot?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

A kite string is 50 m long and makes a 37° angle with the level ground. Use sin 37° ≈ 0.60 and cos 37° ≈ 0.80.

About how high is the kite?

A. 20 m

B. 30 m

C. 40 m

D. 83 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

True or false?

A right triangle with two 45° angles and legs of 7 has a hypotenuse of $7\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

True or false?

$\tan 45° = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 8.G.B.7

FORWARD 2

Use right-triangle reasoning to make design decisions under constraints • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A ramp must rise 2 feet. Building rules allow at most 1 foot of rise for every 12 feet of horizontal run.

What is the shortest allowed run, and how long is the ramp surface, to the nearest tenth of a foot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

A 50-inch TV (measured on the diagonal) is 40 inches wide. A cabinet opening is 32 inches tall and 42 inches wide.

Does the TV fit? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

A cable runs from the top of a 12 m pole to a stake on level ground. Safety rules say the stake must be at least 5 m from the pole, and a path makes 6 m to 8 m from the pole off-limits. The company wants the shortest cable.

Where should the stake go, and how long is the cable?

A. 5 m from the pole; 13 m of cable

B. 9 m from the pole; 15 m of cable

C. At the base of the pole; 12 m of cable

D. 12 m from the pole; about 17 m of cable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

True or false?

A rectangular field is 120 m by 50 m. A diagonal path is 40 m shorter than walking along two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

True or false?

A 10-foot ladder must reach a window 9 feet up, and its base must be at least 5 feet from the wall. This ladder meets both rules.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in three dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Identify the faces, edges, and vertices of a prism	**6.G.A.4**
BACKWARD 2	Describe cross sections of right prisms	**7.G.A.3**
FORWARD 1	Use trigonometric ratios and the Pythagorean theorem to solve rig	**HSG-SRT.C.8**
FORWARD 2	Identify solids made by rotating two-dimensional figures — neare	**HSG-GMD.B.4**

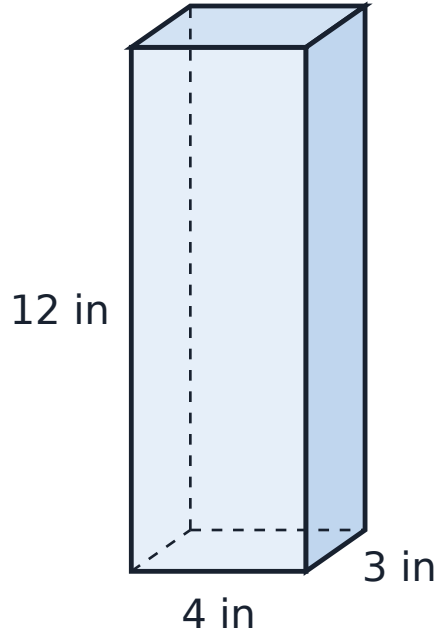
SET 47 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

A box is 4 inches long, 3 inches wide, and 12 inches tall.

How long is the diagonal from a bottom corner to the opposite top corner?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

A cube has 6 cm edges.

Find the diagonal of one face, and then the diagonal from one corner of the cube to the opposite corner. Give exact values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

Will a 26-inch stick fit inside a box that is 24 by 6 by 8 inches, from one corner to the opposite corner?

A. No; the box is too short.

B. Yes; the face diagonal is 30 inches.

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

D. No; the diagonal is only 24 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

A right square pyramid has its top directly above the center of its base. Its base edges are 10 m and its height is 12 m.

What is the slant height (from the top to the midpoint of a base edge)?

Slant height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

True or false?

The corner-to-corner diagonal of a 2 by 3 by 6 box is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

SET 47 • 8.G.B.7

BACKWARD 1

Identify the faces, edges, and vertices of a prism • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

How many edges does a rectangular prism have?

Edges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

How many vertices does a cube have?

A. 6

B. 4

C. 12

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

Each face of a rectangular prism is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A triangular prism has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 8.G.B.7

BACKWARD 2

Describe cross sections of right prisms • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane through two opposite vertical edges.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

A cube is sliced by a plane parallel to one face, through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

A right rectangular prism is sliced parallel to its base, through its interior.

What is the cross section?

A. A circle

B. A triangle

C. A pentagon

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

True or false?

A diagonal slice through a box can make a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A slice of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 8.G.B.7

FORWARD 1

Use trigonometric ratios and the Pythagorean theorem to solve right-triangle problems in three dimensions • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A room is 12 ft long, 9 ft wide, and 8 ft high.

Find the diagonal from a floor corner to the opposite ceiling corner, and the angle it makes with the floor diagonal. Use $\tan \theta = 8 \div 15$ and $\tan 28° \approx 0.53$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

A right square pyramid has base edges of 10 m and a height of 12 m.

Find the angle between a face's slant height and the base, to the nearest degree. Use tan 67° ≈ 2.36 and tan 68° ≈ 2.48.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

A 6 m ladder leans against a wall and makes a 70° angle with the level ground. Use sin 70° ≈ 0.94 and cos 70° ≈ 0.34.

About how high up the wall does it reach?

A. About 5.6 m

B. About 17.6 m

C. About 6.4 m

D. About 2.0 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

True or false?

In a right triangle, if $\tan A = 1$, then angle $A = 45°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

True or false?

The corner-to-corner diagonal of a 3 by 4 by 12 box makes an angle with the floor whose tangent is 12 ÷ 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 8.G.B.7

FORWARD 2

Identify solids made by rotating two-dimensional figures • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

⚑ No high-school standard directly continues this skill at this step. This branch uses the nearest related high-school standard (placed at Grade 10 in this bank).

A right triangle with legs 3 and 4 is rotated around the leg of length 4.

What solid does it make, and what is its slant height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

A 2-by-5 rectangle is rotated around one of its 5-unit sides.

What solid does it make?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

A sphere is sliced by a plane through its center.

What is the cross section?

A. A circle with the same radius as the sphere

B. An oval that is not a circle

C. A square

D. A smaller circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Rotating a semicircle about its diameter makes a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A cube can never have a triangular cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 48 • 8.G.B.8 • GEOMETRY

Find the distance between two points on the coordinate plane

MAIN	Grade 8 main question	**8.G.B.8**
BACKWARD 1	Find distances along grid lines	**6.NS.C.8**
BACKWARD 2	Find a distance as the absolute value of a difference	**7.NS.A.1.c**
FORWARD 1	Use the Pythagorean theorem and trigonometric ratios with coord.	**HSG-SRT.C.8**
FORWARD 2	Use coordinates to find perimeters and areas of polygons	**HSG-GPE.B.7**

SET 48 • 8.G.B.8

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (7, 10).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

Find the distance between $(-3, 4)$ and $(2, -8)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

What is the distance between (0, 0) and (4, 4), to the nearest tenth?

A. 16.0 units

B. 4.0 units

C. 5.7 units

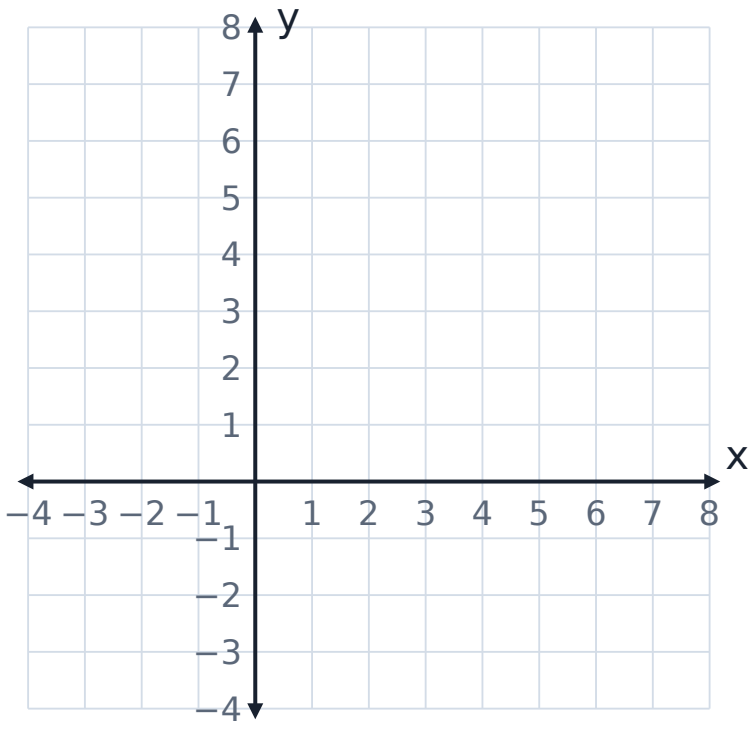
D. 8.0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

Plot A(−2, −1) and B(4, 7). Draw a right triangle with AB as its hypotenuse, and find AB.



AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

True or false?

The distance between $(-1, -1)$ and $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

SET 48 • 8.G.B.8

BACKWARD 1

Find distances along grid lines • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

Which two points are 8 units apart?

A. (1, 1) and (1, 8)

B. (−3, 0) and (3, 0)

C. (−3, 0) and (5, 0)

D. (0, −4) and (0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

True or false?

(2, −5) and (2, 5) are 10 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$(-6, 3)$ and $(-1, 3)$ are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 8.G.B.8

BACKWARD 2

Find a distance as the absolute value of a difference • Grade 7 • The Number System • 7.NS.A.1.c

Find $|7 - (-2)|$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

How far apart are −4 and 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

Which expression gives the distance between -8 and -3?

A. $|-8 - (-3)|$

B. $-8 - 3$

C. $|-8 + 3| - 1$

D. $-8 + (-3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

True or false?

The distance between -5 and 1 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

True or false?

The distance between -7 and -2 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 8.G.B.8

FORWARD 1

Use the Pythagorean theorem and trigonometric ratios with coordinates and distances • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A drone flies in a straight line from (0, 0) to (12, 5) on a grid in meters.

How far does it fly, and what angle does its path make with the x-axis, to the nearest degree? Use tan 22° ≈ 0.40 and tan 23° ≈ 0.42.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

From a point on level ground 40 m from a building, the angle up to the top of the building is 35°. Use tan 35° ≈ 0.70.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

Which point is exactly 5 units from (1, 1)?

A. (7, 1)

B. (5, 5)

C. (1, 7)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

True or false?

The segment from (0, 0) to (3, 3) makes a 45° angle with the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

True or false?

The points (0, 0) and (3, 3) are 6 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 8.G.B.8

FORWARD 2

Use coordinates to find perimeters and areas of polygons • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

Find the area of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

A rhombus has vertices (0, 0), (3, 4), (8, 4), and (5, 0).

What is its perimeter?

A. 20 units

B. 18 units

C. 16 units

D. 25 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

True or false?

A square with vertices (0, 0), (3, 1), (2, 4), and (−1, 3) has an area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

True or false?

The rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 8.G.C.9 • GEOMETRY

Find the volume of a cylinder

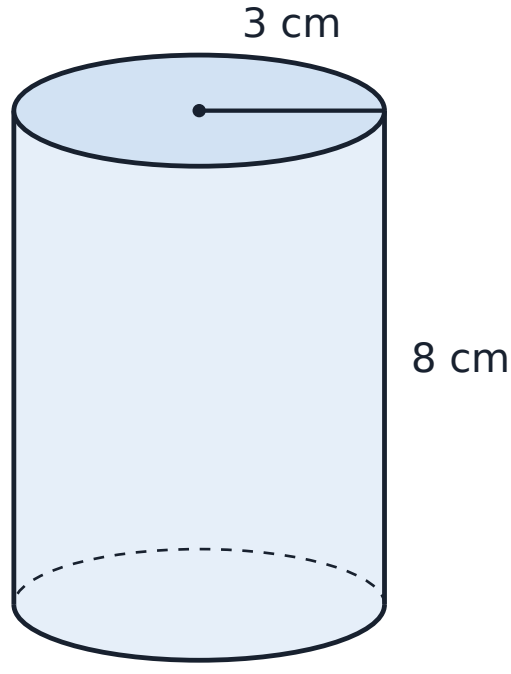
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle	**7.G.B.4**
BACKWARD 2	Find the volume of a right prism as base area times height	**7.G.B.6**
FORWARD 1	Give an informal argument for volume formulas	**HSG-GMD.A.1**
FORWARD 2	Use density with volume in a model	**HSG-MG.A.2**

SET 49 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cylinder in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

A cylinder has a diameter of 10 inches and a height of 6 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

A can holds 128π cubic centimeters and has a radius of 4 cm.

What is its height?

A. 8 cm

B. 2 cm

C. 16 cm

D. 32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

A cylinder-shaped tank has a radius of 2 m and a height of 5 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

True or false?

Doubling the radius of a cylinder while keeping its height the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

SET 49 • 8.G.C.9

BACKWARD 1

Find the area of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 3.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

A circle has a diameter of 10 cm.

What is its area? Use 3.14 for π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

A circle has a radius of 4 m.

What is its area in terms of π?

A. 64π square meters

B. 4π square meters

C. 16π square meters

D. 8π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 1 has an area of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A circle with a diameter of 6 has an area of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism as base area times height • Grade 7 • Geometry • 7.G.B.6

A right prism has a base area of 20 square cm and a height of 7 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

A right triangular prism has a triangle base with area 6 square inches and a length of 10 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

A right prism has a base area of 15 square m and a height of 4 m.

What is its volume?

A. 19 cubic meters

B. 30 cubic meters

C. 225 cubic meters

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

True or false?

The volume of any right prism is the area of its base times its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

A prism with a base area of 9 and a height of 2 has a volume of 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 8.G.C.9

FORWARD 1

Give an informal argument for volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cylinder is the area of its base times its height.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

A straight stack of coins and a slanted stack have the same coins and the same height.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

An oblique cylinder and a right cylinder have the same radius and height.

How do their volumes compare?

A. The oblique one is larger.

B. It cannot be known.

C. The right one is larger.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

True or false?

Cavalieri's principle compares the areas of cross sections at every height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Leaning a stack of paper to one side increases its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 8.G.C.9

FORWARD 2

Use density with volume in a model • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum cylinder has a radius of 2 cm and a height of 10 cm. Aluminum has a density of 2.7 grams per cubic centimeter.

What is its mass? Use 3.14 for π.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

A town has 12,000 people and covers 30 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

A 500 cubic-centimeter block has a mass of 4,000 grams.

What is its density?

A. 8 grams per cubic centimeter

B. 2 grams per cubic centimeter

C. 0.125 gram per cubic centimeter

D. 4,500 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

True or false?

If two objects have the same volume, the one with the greater density has the greater mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

True or false?

Density is volume divided by mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 8.G.C.9 • GEOMETRY

Find the volume of a cone

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle from its radius or diameter	**7.G.B.4**
BACKWARD 2	Multiply a fraction by a whole number	**5.NF.B.4.a**
FORWARD 1	Explain why a cone's volume is one-third of a cylinder's	**HSG-GMD.A.1**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

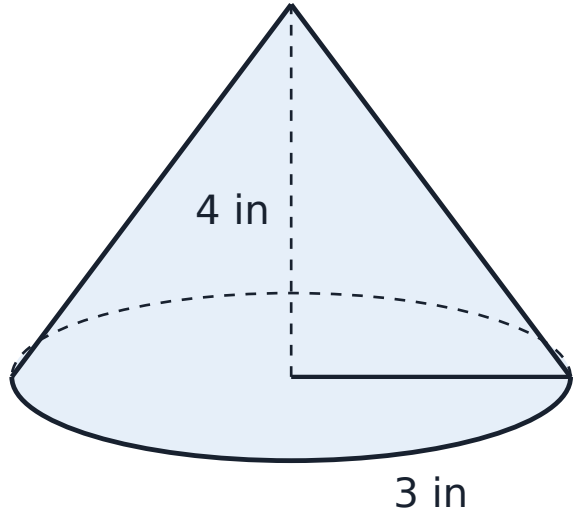
SET 50 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cone in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

A cone has a diameter of 12 feet and a height of 10 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. The cylinder holds 90 cubic cm.

How much does the cone hold?

A. 30 cubic cm

B. 90 cubic cm

C. 270 cubic cm

D. 45 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

A cone has a volume of 50π cubic units and a radius of 5 units.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

True or false?

A cone with a radius of 6 and a height of 2 has a volume of 24π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

SET 50 • 8.G.C.9

BACKWARD 1

Find the area of a circle from its radius or diameter • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 6.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

A circle has a diameter of 4 m.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

A circle has a radius of 10 cm.

What is its area? Use 3.14 for π.

A. 314 square cm

B. 31.4 square cm

C. 62.8 square cm

D. 628 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

True or false?

A circle with a radius of 5 has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 8.G.C.9

BACKWARD 2

Multiply a fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{1}{3} \times 36$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

Multiply.

$\frac{1}{3} \times 90$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

Multiply.

$\frac{1}{3} \times 48$

A. 144

B. 12

C. 16

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{3} \times 27 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 15 = 45$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 8.G.C.9

FORWARD 1

Explain why a cone's volume is one-third of a cylinder's • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cone and a cylinder have the same base and height. Water from 3 full cones exactly fills the cylinder.

What formula does this suggest for the cone's volume?

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

Explain why an oblique cone and a right cone with the same base and height have the same volume.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

A pyramid and a prism have the same base and height.

How do their volumes compare?

A. The pyramid's is $\frac{1}{3}$ of the prism's.

B. The prism's is $\frac{1}{3}$ of the pyramid's.

C. The pyramid's is $\frac{1}{2}$ of the prism's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

True or false?

A cone's volume formula has the same form as a pyramid's: $\frac{1}{3} \times$ base area $\times$ height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

True or false?

A cone has half the volume of a cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A paper cone cup has a radius of 4 cm and a height of 9 cm.

How many full cups fill a cylinder-shaped jug with a radius of 8 cm and a height of 18 cm?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

A cone-shaped pile of sand has a radius of 3 m and holds 30π cubic meters.

How tall is the pile?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

A right circular cone has a radius of 6 and a slant height of 10.

What is its volume? (Find its height first.)

A. 120π

B. 96π

C. 32π

D. 288π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

True or false?

Doubling the height of a cone while keeping its radius the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

True or false?

Doubling the radius of a cone while keeping its height the same doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 8.G.C.9 • GEOMETRY

Find the volume of a sphere

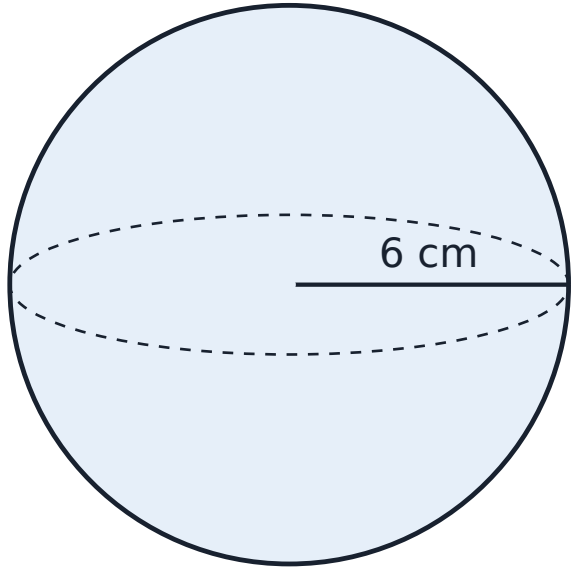
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a cube	**6.G.A.2**
BACKWARD 2	Evaluate a formula with a fraction and an exponent	**6.EE.A.2.c**
FORWARD 1	Model objects with spheres, cylinders, and cones	**HSG-MG.A.1**
FORWARD 2	Use volume formulas to solve problems with spheres	**HSG-GMD.A.3**

SET 51 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the sphere in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

A ball has a diameter of 10 cm.

Find its volume to the nearest tenth. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

A hemisphere (half of a sphere) has a radius of 3.

What is its volume in terms of π?

A. 36π

B. 18π

C. 9π

D. 27π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

A sphere has a volume of 36π cubic units.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

True or false?

A sphere with a radius of 1 has a volume of $\frac{4}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

SET 51 • 8.G.C.9

BACKWARD 1

Find the volume of a cube • Grade 6 • Geometry • 6.G.A.2

A cube has edges of 3 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

A cube has edges of $\frac{1}{2}$ foot.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

A cube has edges of 6 cm.

What is its volume?

A. 18 cubic cm

B. 216 cubic cm

C. 72 cubic cm

D. 36 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

True or false?

A cube with 5-meter edges has a volume of 15 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 8.G.C.9

BACKWARD 2

Evaluate a formula with a fraction and an exponent • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $V = \frac{4}{3}r^3$. Find V when r = 3.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

Use $V = \frac{4}{3}r^3$. Find V when r = 6.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

Use $V = \frac{1}{2}r^3$. Find V when r = 2.

A. 6

B. 4

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

True or false?

When $r = 1$, $\frac{4}{3}r^3 = \frac{4}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

True or false?

When $r = 2$, $\frac{4}{3}r^3 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 8.G.C.9

FORWARD 1

Model objects with spheres, cylinders, and cones • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

An orange is modeled as a sphere with a diameter of 8 cm.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

A tennis ball can holds 3 balls in a cylinder whose radius equals the ball's radius r and whose height is 6r.

What fraction of the can is filled by the balls?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

Which shape is the best model for a scoop of ice cream?

A. A cylinder

B. A sphere

C. A cone

D. A cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

True or false?

Modeling Earth as a sphere is a useful approximation even though Earth is slightly flattened.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

True or false?

A soccer ball is best modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems with spheres • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A spherical tank has a radius of 3 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r.

What is the ratio of the sphere's volume to the cylinder's volume?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

Doubling a sphere's radius multiplies its volume by...

A. 6

B. 2

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

True or false?

A sphere with volume 288π has a radius of 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

True or false?

A hemisphere with radius 6 has a volume of 288π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 8.G.C.9 • GEOMETRY

Solve real-world problems with the volumes of cylinders, cones, and spheres

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a solid made of two prisms	**5.MD.C.5.c**
BACKWARD 2	Find the volume of a right prism	**7.G.B.6**
FORWARD 1	Model real objects with combinations of solids	**HSG-MG.A.1**
FORWARD 2	Apply volume to design and capacity problems	**HSG-MG.A.3**

SET 52 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

A grain silo is a cylinder with a radius of 6 m and a height of 10 m, topped with half of a sphere with a radius of 6 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

An ice cream cone (radius 3 cm, height 10 cm) is filled and topped with half of a sphere of ice cream with a radius of 3 cm.

Find the total volume of ice cream in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

A cone and a cylinder each have a radius of 2 and a height of 6.

How many full cones of water fill the cylinder?

A. 6

B. 3

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

A straight-sided cylindrical container has an inside radius of 5 cm and holds water. It is tall enough that no water spills. A ball with a radius of 3 cm is dropped in and sinks completely under the water.

How many centimeters does the water level rise?

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

True or false?

A sphere with radius r fits exactly in a cylinder with radius r and height 2r. The sphere's volume is $\frac{2}{3}$ of the cylinder's.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

SET 52 • 8.G.C.9

BACKWARD 1

Find the volume of a solid made of two prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 4 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

Two cubes with 2-unit edges are joined face to face.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

A 5 by 2 by 1 prism and a 1 by 2 by 1 prism are joined with no overlap.

What is the total volume?

A. 12 cubic units

B. 7 cubic units

C. 20 cubic units

D. 10 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

Volumes of parts that do not overlap can be added to find a total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

A 3 by 3 by 3 cube and a 1 by 1 by 1 cube joined with no overlap have a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism • Grade 7 • Geometry • 7.G.B.6

A right rectangular prism is 6 cm by 4 cm by 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

A right prism has a base area of 12 square feet and a height of 3 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

A right triangular prism has a base area of 10 square inches and a length of 8 inches.

What is its volume?

A. 40 cubic inches

B. 80 cubic inches

C. 18 cubic inches

D. 800 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A cube with edges of 10 m has a volume of 1,000 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A 2 by 5 by 10 box has a volume of 17 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 8.G.C.9

FORWARD 1

Model real objects with combinations of solids • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A capsule-shaped vitamin is a cylinder 1 cm long with a radius of 0.3 cm, with half of a sphere of radius 0.3 cm on each end.

What solids make up the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

A pencil is modeled as a cylinder with a cone at one end.

Which measurements are needed to find its volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

Which solids best model a traffic cone sitting on a square base?

A. A cone on a square prism

B. A sphere on a cylinder

C. Two cylinders

D. A cube on a pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

True or false?

A model can use simpler shapes even when the real object is not exactly that shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

True or false?

A water bottle is best modeled as one sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

SET 52 • 8.G.C.9

FORWARD 2

Apply volume to design and capacity problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a cylinder-shaped can that holds 500π cubic cm with a radius of 5 cm.

How tall must the can be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

A cone-shaped funnel has a radius of 6 cm and a height of 10 cm. Water drains at 6π cubic cm per second.

How long does a full funnel take to drain?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

Each can is a cylinder holding the same volume.

Which design uses the most metal for the curved side?

A. Radius 4 cm, height 6.25 cm

B. Radius 2.5 cm, height 16 cm

C. Radius 5 cm, height 4 cm

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

True or false?

Two cans with the same volume can have different surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

True or false?

A taller can always holds more than a shorter can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 8.SP.A.1 • STATISTICS & PROBABILITY

Construct a scatter plot for bivariate measurement data

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Graph points that represent a real-world situation	**5.G.A.2**
BACKWARD 2	Display a numerical data set in a dot plot	**6.SP.B.4**
FORWARD 1	Compare linear, quadratic, and exponential models for two-variabl…	**HSS-ID.B.6**
FORWARD 2	Fit an exponential or quadratic function to data and use it to make	**HSS-ID.B.6.a**

SET 53 • 8.SP.A.1

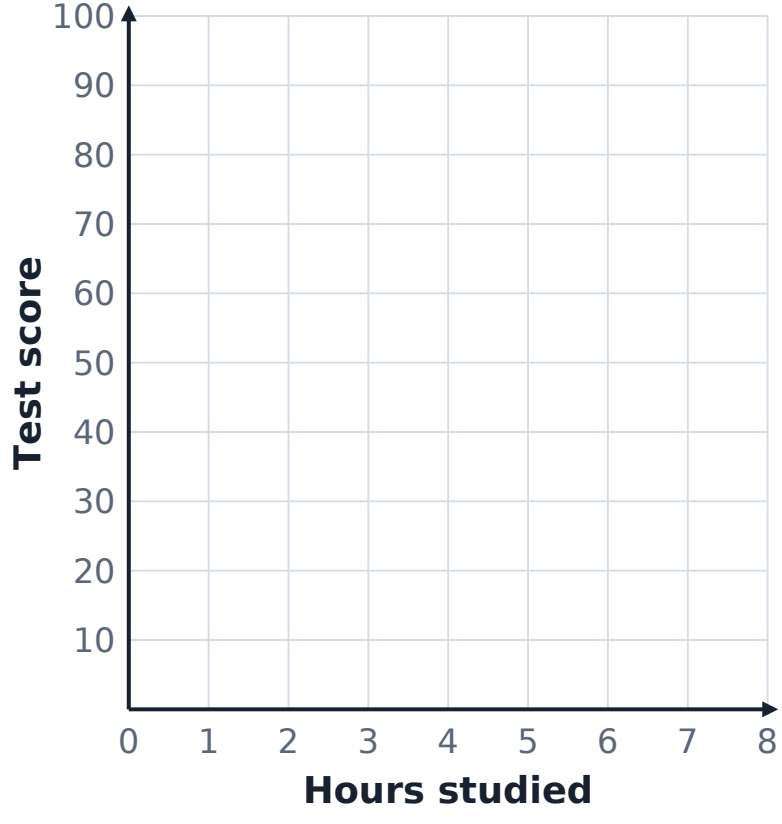
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Seven students recorded (hours studied, test score):

(1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), (7, 90)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

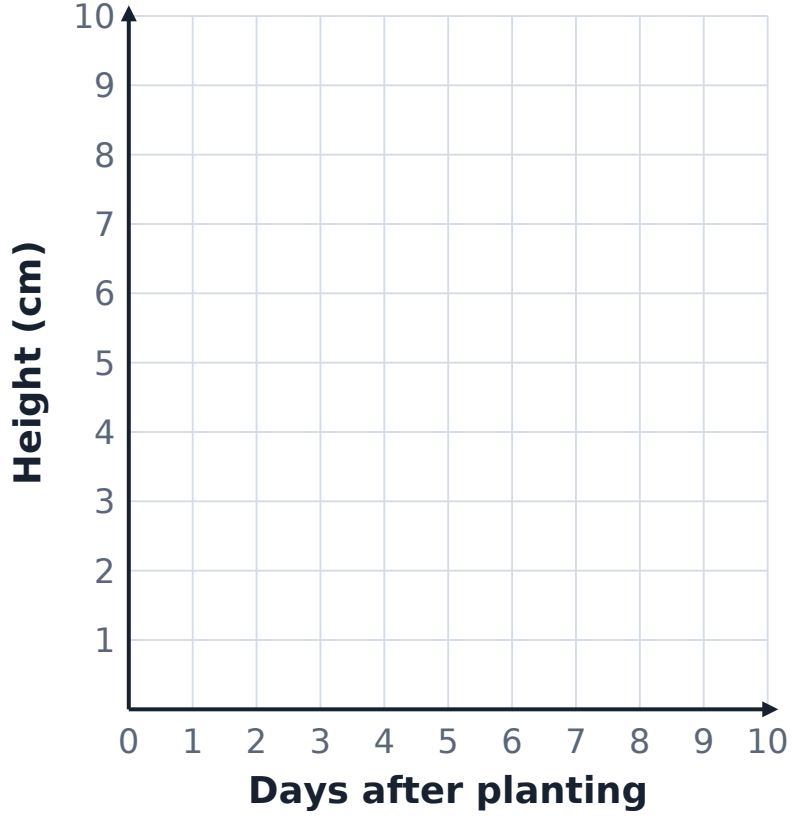
ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

A bean plant was measured on different days (days after planting, height in cm):

(2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), (10, 8)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

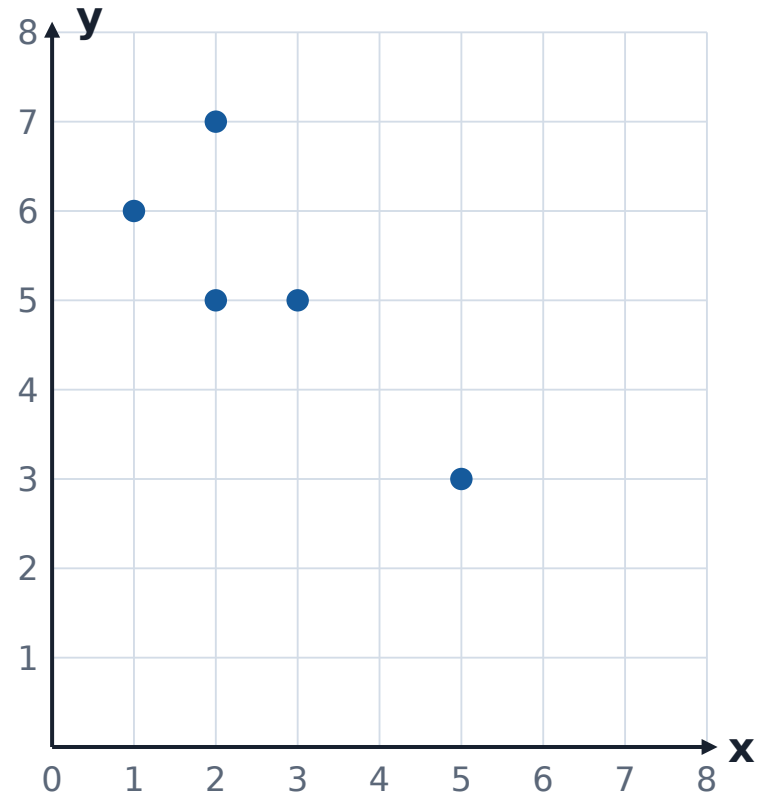
ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

The scatter plot should show the data (1, 6), (2, 5), (3, 5), (5, 3), and (7, 2), but one point is plotted wrong.

Which point is wrong, and where should it be?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

Ten students recorded their shoe sizes and heights to make a scatter plot.

What does each point on the scatter plot show?

A. How many students have each shoe size

B. One student's shoe size and height

C. The mean shoe size of the class

D. One student's shoe size only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

True or false?

A scatter plot of hours slept and test scores puts two points on the grid for each student, one for each variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

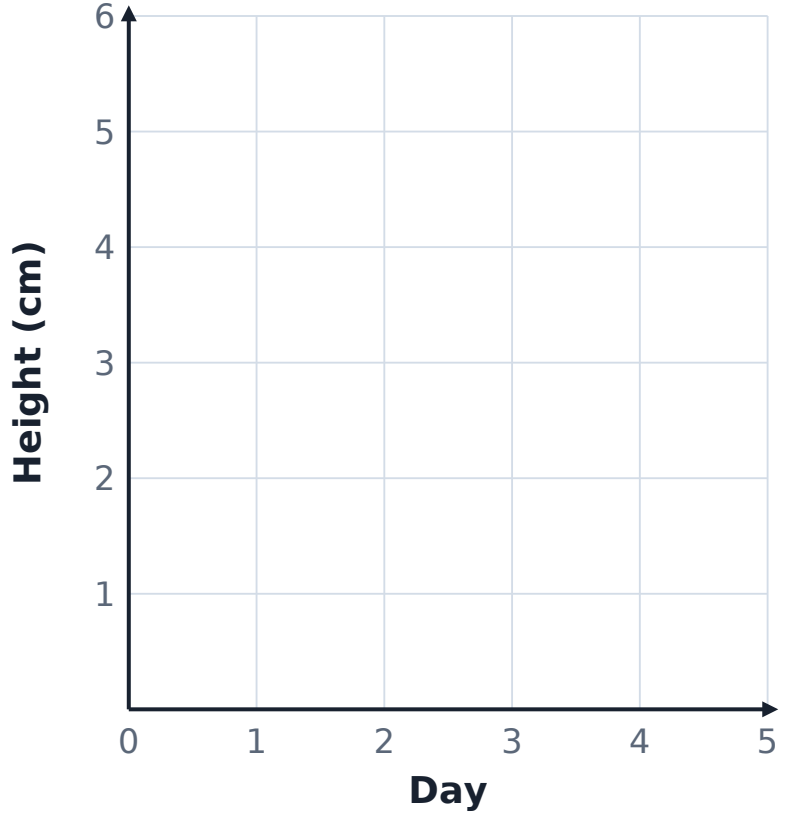
SET 53 • 8.SP.A.1

BACKWARD 1

Graph points that represent a real-world situation • Grade 5 • Geometry • 5.G.A.2

A plant was 2 cm tall on day 1, 4 cm tall on day 2, and 5 cm tall on day 3.

Plot the points (day, height) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

A graph shows points (day, height in cm).

What does the point (4, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

Points on a graph are (hours, miles).

Which point shows 5 hours and 2 miles?

A. (5, 5)

B. (2, 5)

C. (5, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

True or false?

To plot (3, 6), start at the origin, move 3 units right, then 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The point (0, 4) is on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 8.SP.A.1

BACKWARD 2

Display a numerical data set in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The numbers of pets of 10 students are 0, 1, 1, 2, 2, 2, 3, 3, 4, and 1.

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

How many students are shown in the dot plot?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

Which display shows each value of a data set as a dot above a number line?

A. A box plot

B. A bar graph

C. A histogram

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

In a dot plot, a value that appears 3 times has 3 dots stacked above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A dot plot shows two different measurements for each student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 1

Compare linear, quadratic, and exponential models for two-variable data, using technology • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Data: (0, 3), (1, 6), (2, 12), (3, 24).

Does a linear or an exponential model fit these data better? Explain using differences and ratios, and give the model.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

Use a calculator or spreadsheet to find the least-squares line for the data (1, 2), (2, 3), (3, 5), (4, 6). Round the slope and the y-intercept to the nearest tenth.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

Which model fits the data (1, 1), (2, 4), (3, 9), (4, 16) exactly?

A. $y = x^2$

B. $y = 2^x$

C. $y = 5x - 5$

D. $y = 4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

True or false?

A scatter plot that curves upward more and more steeply may be fit better by an exponential or quadratic function than by a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

True or false?

A line with residuals 0, 3, −5, and 4 fits the same data better than a curve with residuals 0, 0.1, −0.2, and 0.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 2

Fit an exponential or quadratic function to data and use it to make predictions • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A bacteria count is 100 at hour 0, 200 at hour 1, 400 at hour 2, and 800 at hour 3.

Write a function that fits the data.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

A bacteria count fits y = 100 · 2^x after x hours.

Predict the count at hour 6.

Count: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

Which function fits the data exactly?

x	0	1	2	3	4
y	0	1	4	9	16

A. $y = x^2$

B. $y = 4x$

C. $y = x + 4$

D. $y = 2^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The data (0, 5), (1, 10), (2, 20), and (3, 40) fit $y = 5 \cdot 2^x$ exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q4/5

True or false?

Data whose y-values go up by the same amount at each step are best fit by an exponential function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

SET 54 • 8.SP.A.1 • STATISTICS & PROBABILITY

Describe clustering, outliers, and positive, negative, linear, or nonlinear association in a scatter plot

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Describe a data distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Describe the overall pattern of data and any striking deviations	**6.SP.B.5.c**
FORWARD 1	Distinguish between correlation and causation	**HSS-ID.C.9**
FORWARD 2	Interpret the correlation coefficient of a linear fit	**HSS-ID.C.8**

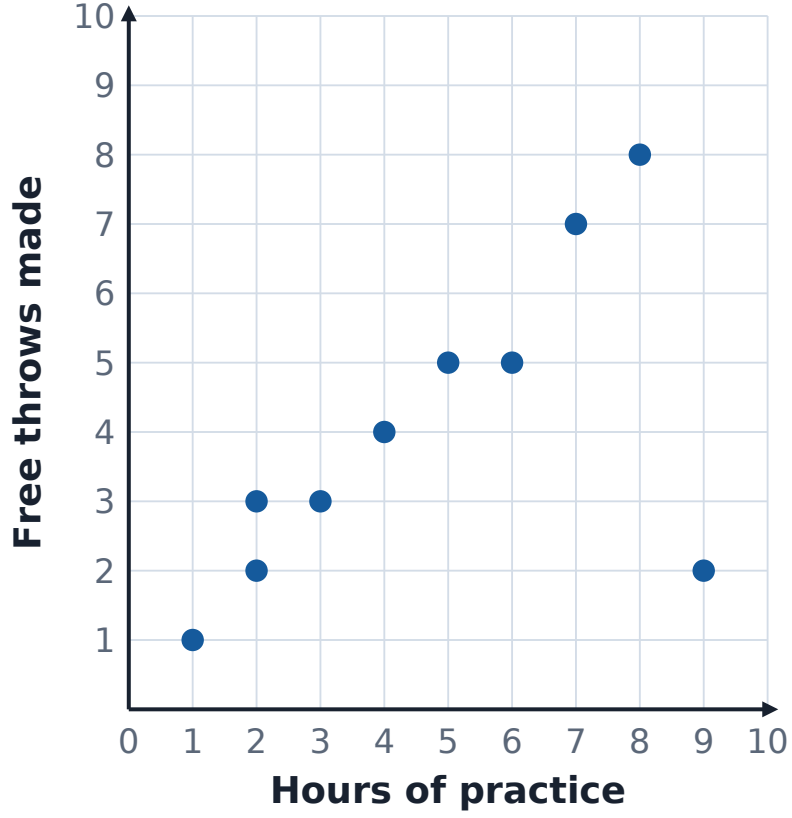
SET 54 • 8.SP.A.1

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Describe the association in the scatter plot (positive, negative, or none; linear or nonlinear), and name any outlier.

Answer: _____



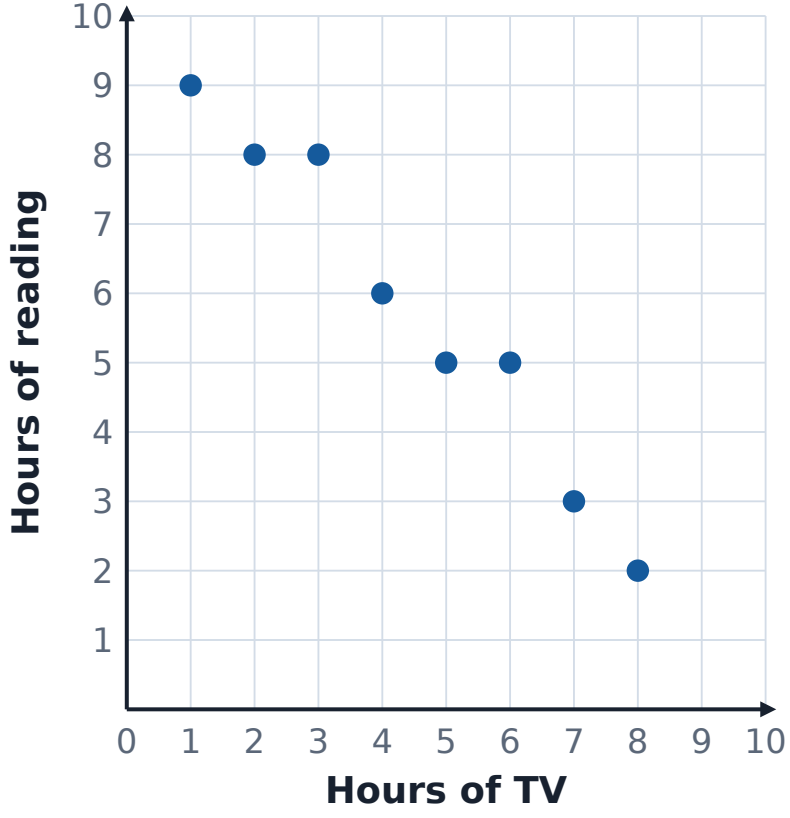
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

Describe the association in the scatter plot.

Answer: _____

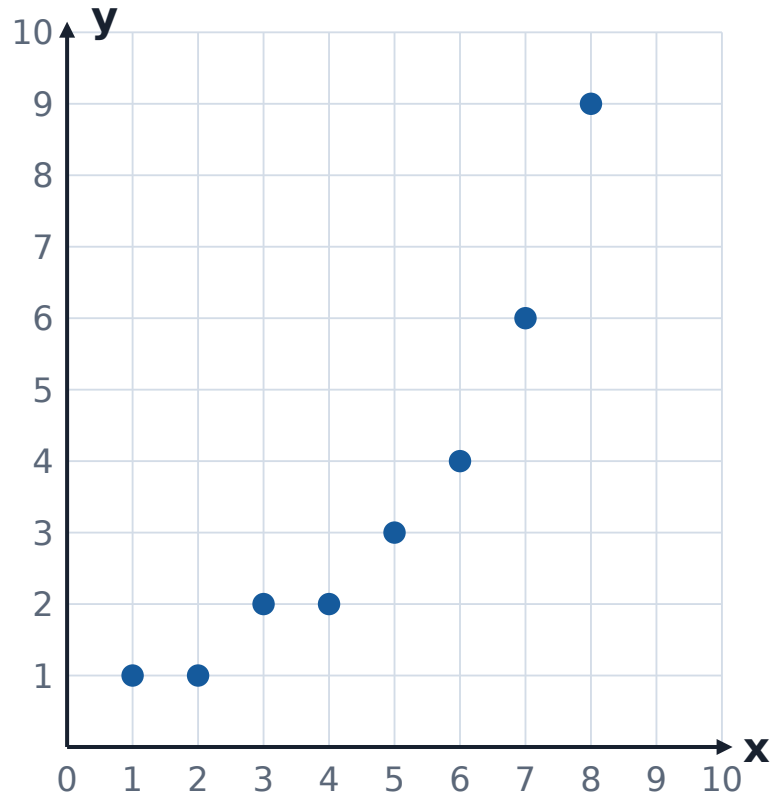


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

Which best describes the association in the scatter plot?



A. Positive and linear

B. No association

C. Negative and nonlinear

D. Positive and nonlinear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

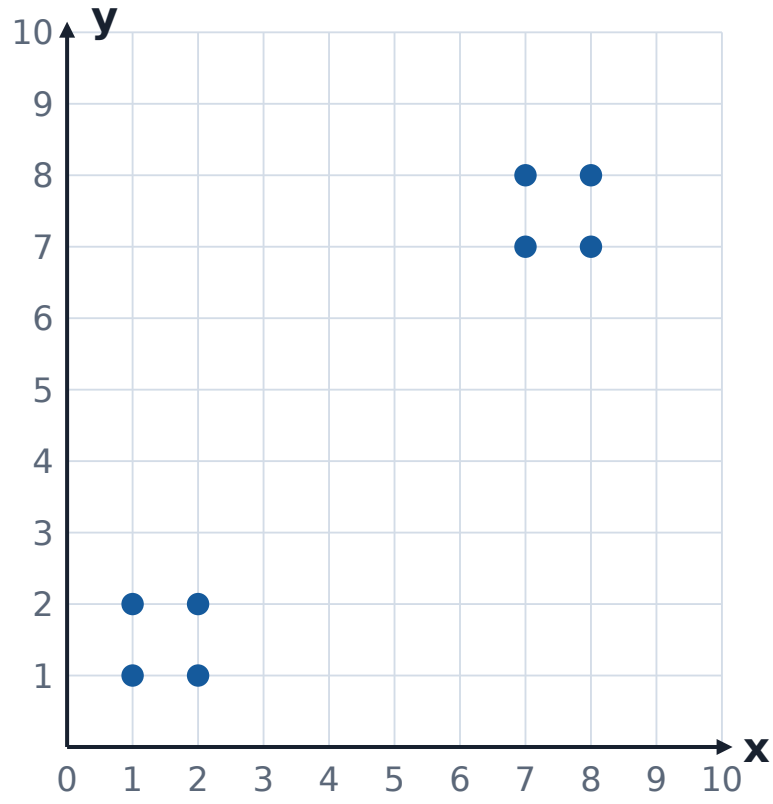
ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

The points in the scatter plot form two groups.

Describe the two clusters.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

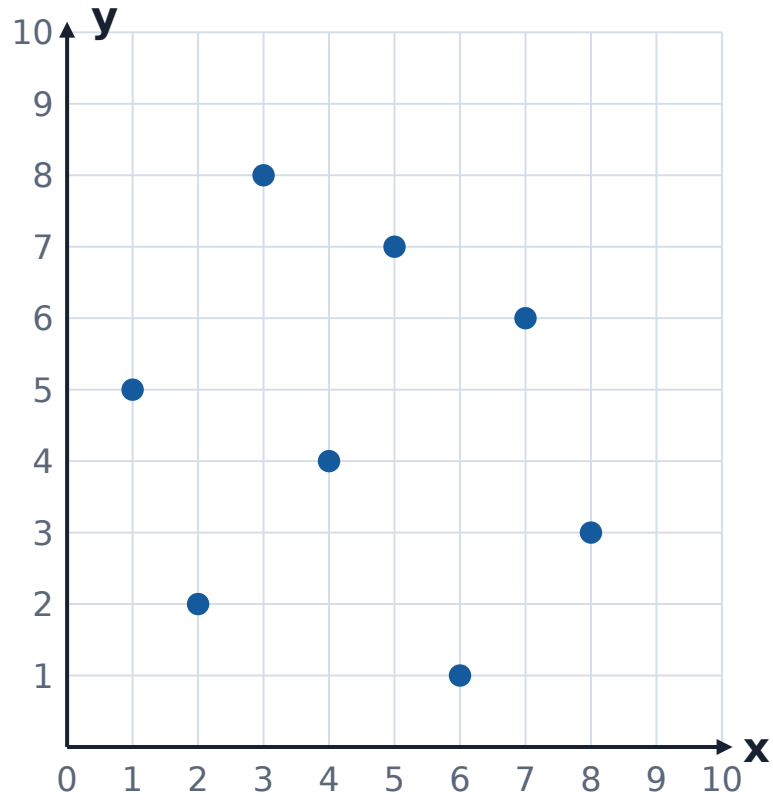
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

True or false?

The scatter plot shows a negative association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

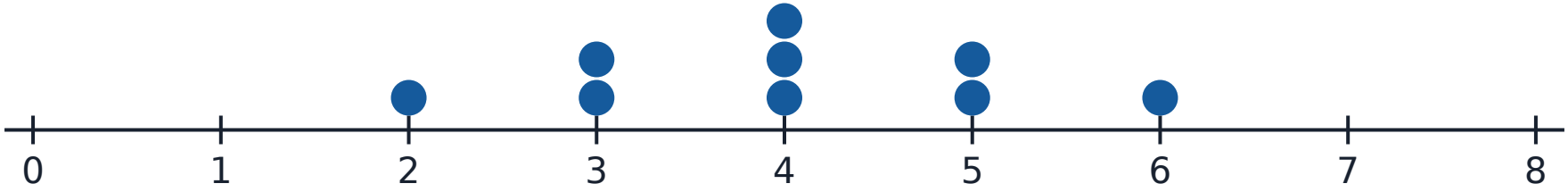
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 1

Describe a data distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Is the data in the dot plot symmetric or skewed?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

What is the range of the data in the dot plot?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which value is far from the rest of the data?

3, 4, 4, 5, 5, 6, 18

A. 18

B. 6

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

The data set 2, 3, 3, 4, 4, 4, 5, 5, 6 is symmetric about 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

The range of a data set measures its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 2

Describe the overall pattern of data and any striking deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

4, 6, 7, 7, 8, 9, 30

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

Which value is a striking deviation from the pattern?

4, 6, 7, 7, 8, 9, 30

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

The data are 10, 12, 12, 13, 14, and 40.

Which measure of center is changed more by the value 40?

A. The mean

B. The median

C. Both are changed equally

D. Neither is changed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

True or false?

Removing a value that is much larger than the others lowers the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

True or false?

An outlier always changes the median by a large amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 1

Distinguish between correlation and causation • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Data show that cities with more ice-cream sales also have more sunburns.

Do these data show that buying ice cream causes sunburns? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

A study finds that students who sleep more tend to have higher grades.

Can you conclude that more sleep causes higher grades? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

Which kind of study can show that one variable causes a change in another?

A. An observational study

B. A randomized experiment

C. A survey of volunteers

D. A scatter plot of survey data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

True or false?

Two variables can be strongly associated without one causing the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

True or false?

A strong correlation shows that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 2

Interpret the correlation coefficient of a linear fit • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use a calculator or spreadsheet to find the correlation coefficient r for the data (1, 2), (2, 4), (3, 5), (4, 4), (5, 6). Round to the nearest hundredth, and describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

Which shows the stronger linear association, $r = 0.45$ or $r = -0.81$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

Points lie almost exactly on a line that rises from left to right.

Which value of r fits?

A. -0.98

B. 0.98

C. 0

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

True or false?

The correlation coefficient r is always between −1 and 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

True or false?

$r = -0.9$ shows a weaker linear association than $r = 0.5$.

A. True

B. False

SET 55 • 8.SP.A.2 • STATISTICS & PROBABILITY

Informally fit a straight line to a scatter plot and judge how closely it fits

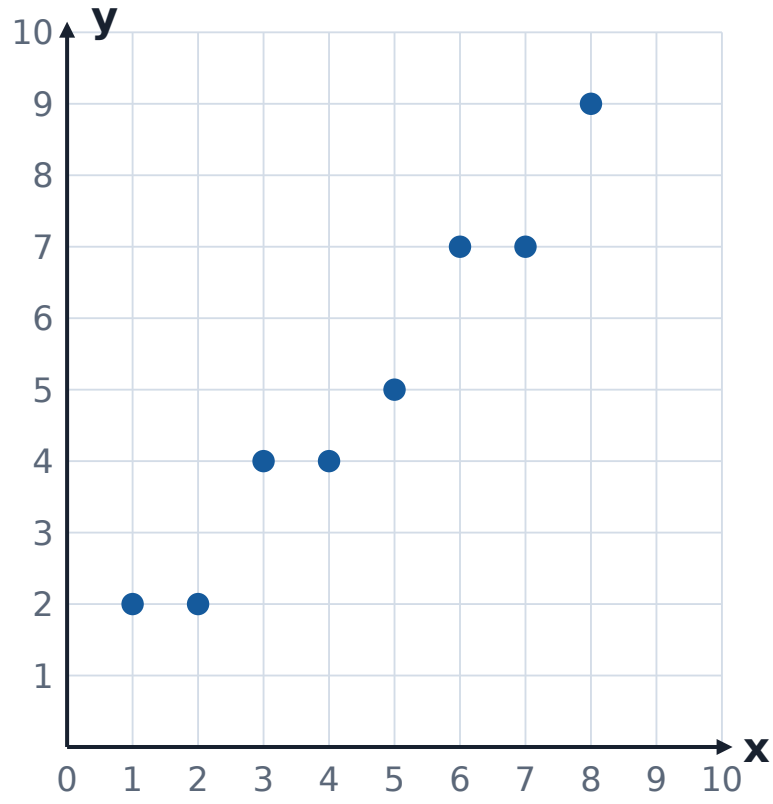
MAIN	Grade 8 main question	**8.SP.A.2**
BACKWARD 1	Decide from a table or graph whether a relationship is proportion…	**7.RP.A.2.a**
BACKWARD 2	Write an equation for a relationship shown in a table or graph	**6.EE.C.9**
FORWARD 1	Fit a least-squares line with technology and compare candidate lin…	**HSS-ID.B.6.c**
FORWARD 2	Judge the fit of a line by finding and plotting residuals	**HSS-ID.B.6.b**

SET 55 • 8.SP.A.2

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.2

Draw a straight line that fits the data. Then write an equation of your line.



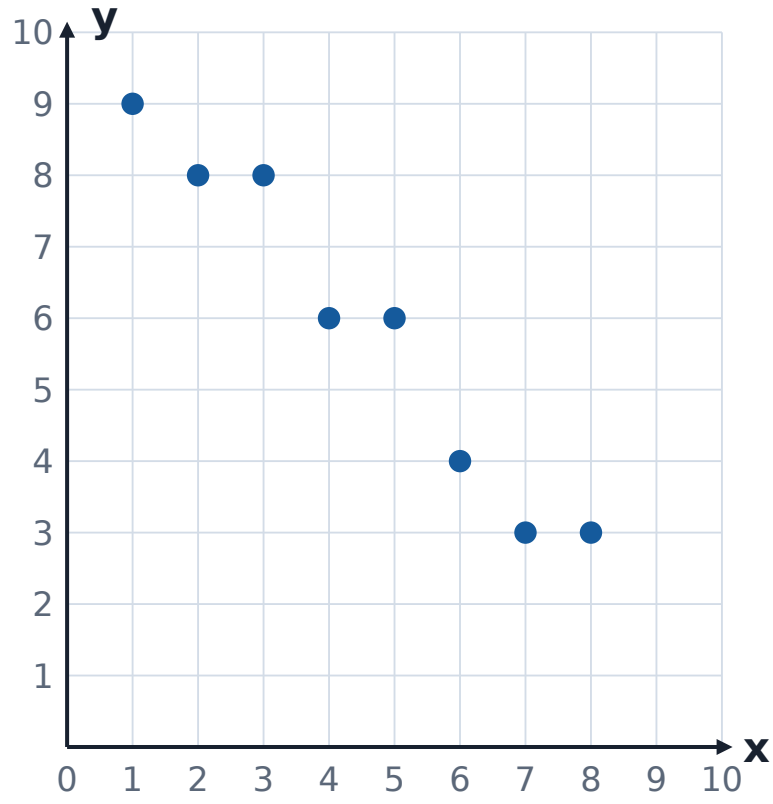
Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

Draw a straight line that fits the data. Use your line to estimate y when x = 9.



Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

Two students drew lines for the same scatter plot. Line A has every point above it. Line B has about half the points above it and half below, all close to it.

Which line fits better? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

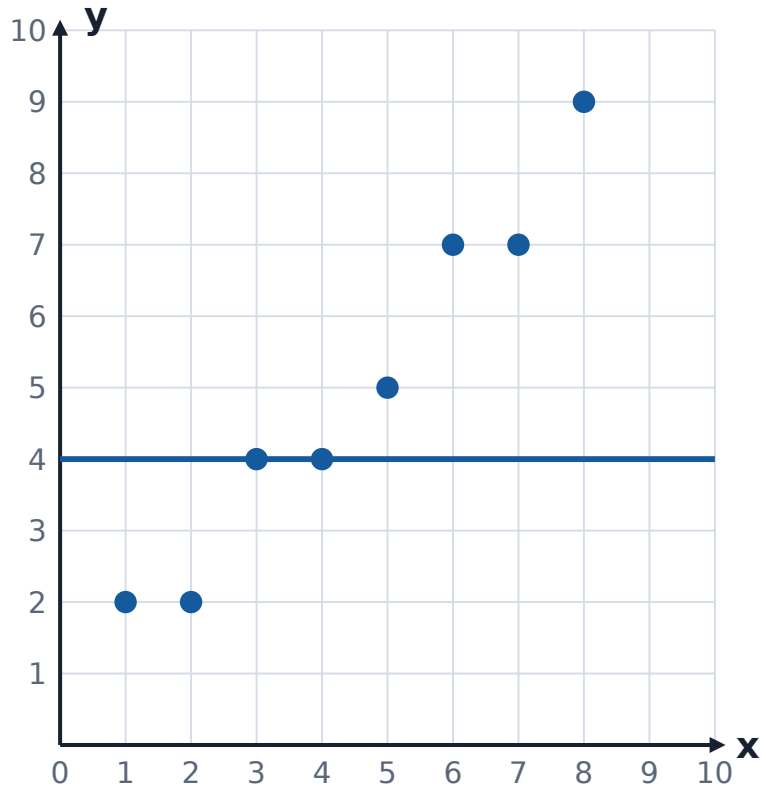
Why is the line on the scatter plot a poor fit?

A. It is not long enough.

B. It does not follow the upward trend of the points.

C. It starts at the y-axis.

D. It passes through too many points.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

True or false?

A line that fits a scatter plot well must pass through every point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

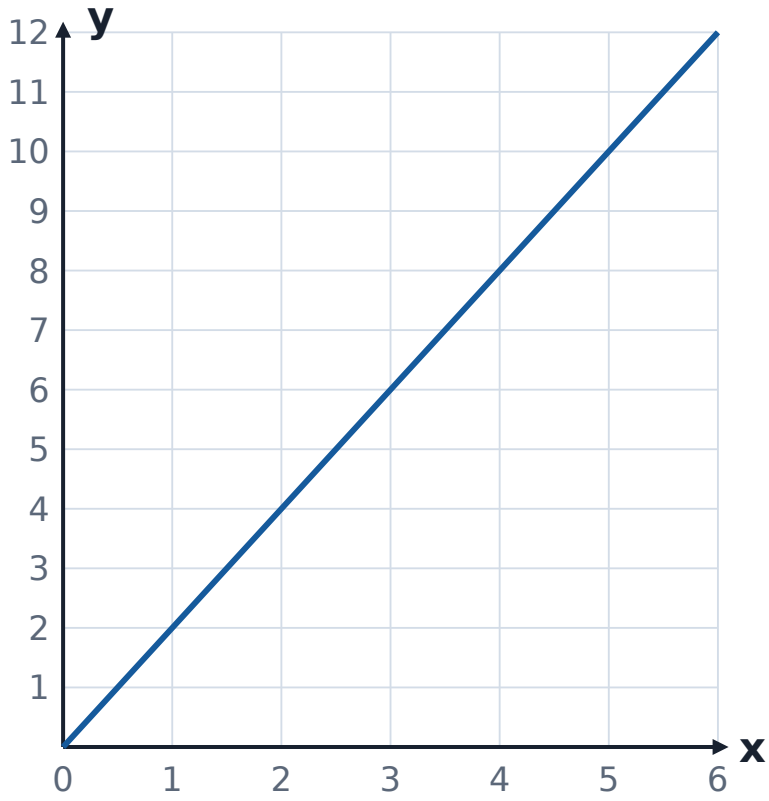
SET 55 • 8.SP.A.2

BACKWARD 1

Decide from a table or graph whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Is the relationship shown in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	4
y	3	6	13

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

Which graph shows a proportional relationship?

A. A curve through (0, 0) and (2, 4)

B. A straight line through (0, 2) and (2, 6)

C. A horizontal line through (0, 3)

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

True or false?

A straight line through (0, 0) and (3, 9) shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

True or false?

Every straight-line graph shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 2

Write an equation for a relationship shown in a table or graph • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	0	1	2	3
y	2	3	4	5

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

A line passes through (0, 0), (1, 5), and (2, 10).

Which equation does it show?

A. $y = 5x$

B. $y = 10x$

C. $x = 5y$

D. $y = x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

True or false?

The points (1, 3), (2, 6), and (3, 9) fit y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

True or false?

The points (0, 1), (1, 2), and (2, 3) fit $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 1

Fit a least-squares line with technology and compare candidate lines • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Use a calculator or spreadsheet to find the least-squares line for the data (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), (6, 11). Round the slope and the y-intercept to the nearest tenth.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

Two lines are proposed for the data (1, 3), (2, 5), (3, 6), (4, 8): $y = 2x + 1$ and $y = 1.6x + 1.5$.

Find the sum of the squared residuals for each line. Which line fits better?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

A least-squares line for hours studied x and test score y is $y = 4.2x + 61$. A student who studied 0 hours scored 70.

What is that student's residual?

A. 70

B. 9

C. 61

D. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

True or false?

The least-squares line makes the sum of the squared residuals as small as possible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

True or false?

A line of fit should always be drawn through the first and last data points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 2

Judge the fit of a line by finding and plotting residuals • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

A line of fit is $y = 2x + 1$. A data point is $(3, 8)$.

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

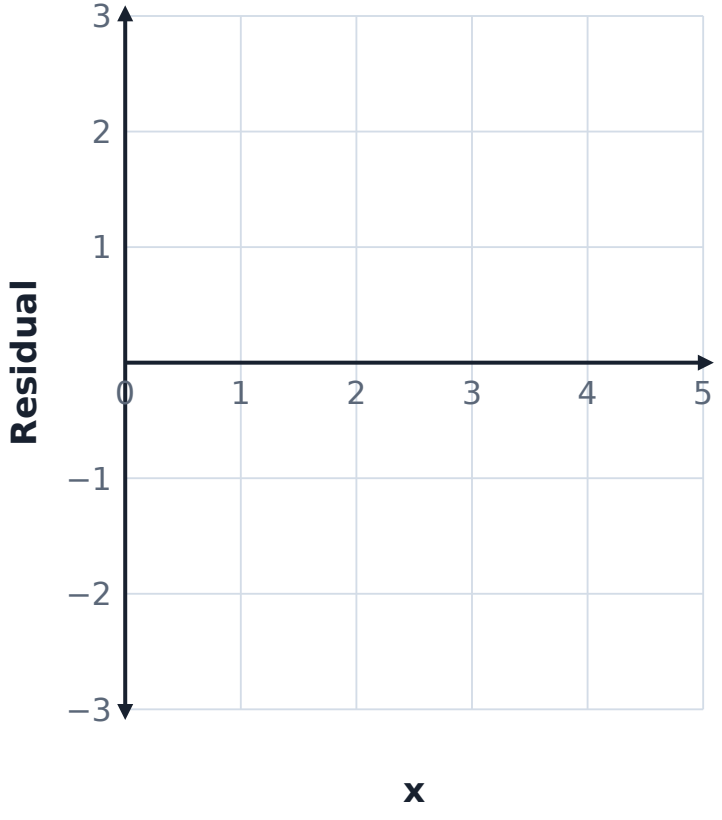
ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

A line of fit is y = 2x + 1 for the data (1, 4), (2, 4), (3, 8), and (4, 9).

Find each residual (actual y minus predicted y) and plot the residuals against x on the grid.

Residuals: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

A residual plot shows small residuals scattered above and below 0 with no pattern.

What does this suggest?

A. A curve must be used.

B. Every prediction is exact.

C. A linear model is a poor choice.

D. A linear model is appropriate for the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

True or false?

A residual of 0 means the data point lies on the line of fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that a linear model fits well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

SET 56 • 8.SP.A.3 • STATISTICS & PROBABILITY

Use the equation of a linear model to interpret the slope and intercept and to make predictions

MAIN	Grade 8 main question	**8.SP.A.3**
BACKWARD 1	Explain what a point on the graph of a proportional relationship m…	**7.RP.A.2.d**
BACKWARD 2	Evaluate a formula for given values of its variable	**6.EE.A.2.c**
FORWARD 1	Interpret the slope and intercept of a linear model in context	**HSS-ID.C.7**
FORWARD 2	Evaluate reports and predictions based on data	**HSS-IC.B.6**

SET 56 • 8.SP.A.3

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.3

A line of fit for the height h (inches) of a plant after w weeks is h = 1.5w + 4.

What do the slope and the intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

A line of fit for the gas g (gallons) left in a car's tank after m miles is $g = 14 - 0.04m$.

Predict the gas left after 200 miles.

Gas left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

A model for the water level w (feet) of a pond t days after the rain stops is $w = 12 - 0.3t$.

What does -0.3 mean?

A. The level rises about 0.3 foot each day.

B. The pond is empty after 0.3 day.

C. The level starts at 0.3 foot.

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

A model for the score s after h hours of practice is $s = 6h + 58$.

Predict the score after 5 hours of practice.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

True or false?

In the model $y = 3.2x + 15$ for the cost of a taxi ride of x miles, 15 is the cost per mile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 1

Explain what a point on the graph of a proportional relationship means • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph of cost (dollars) against pounds is proportional and passes through (1, 3).

What does (1, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

A proportional graph of miles against hours passes through (0, 0).

What does (0, 0) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

A proportional graph of (gallons, cost) passes through (1, 4).

What is the unit rate?

A. $5 per gallon

B. $4 per gallon

C. $0.25 per gallon

D. $1 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

True or false?

On a proportional graph of (hours, pages), the point (1, 20) means 20 pages are read per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

True or false?

On a proportional graph of (tickets, cost), the point (2, 18) means that one ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 2

Evaluate a formula for given values of its variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate 1.5w + 4 when w = 6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

Evaluate $300 - 40t$ when $t = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

Evaluate 6h + 58 when h = 2.

A. 62

B. 128

C. 66

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

True or false?

When $x = 4$, $3x + 5 = 35$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

True or false?

When $t = 2$, $10 - 3t = 14$.

A. True

B. False

SET 56 • 8.SP.A.3

FORWARD 1

Interpret the slope and intercept of a linear model in context • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for the number n of ice pops sold on a day with a high temperature of t °F is $n = 4t - 200$.

What does the slope mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

A model for ice-pop sales is $n = 4t - 200$, where t is the temperature in °F.

Does the intercept -200 make sense in this situation? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

A model for a phone's battery level b (percent) after h hours of use is $b = 100 - 12.5h$.

What does 100 mean?

A. The battery lasts 100 hours.

B. The battery charges 100% each hour.

C. The battery loses 100% each hour.

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

True or false?

In the model $y = 0.8x + 2$, y increases by about 0.8 for each increase of 1 in x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

True or false?

In a linear model, the intercept always has a sensible meaning in the situation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 2

Evaluate reports and predictions based on data • High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A model of children's heights for ages 2 to 10 is $h = 2.5a + 31$ inches at age a.

A report uses it to predict the height of a 40-year-old. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

A line of fit was made from data collected between 60 °F and 90 °F, and the data points lie close to the line.

A report uses it to predict sales at 70 °F. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

A report says: "People who own more books live longer, so buying books makes you live longer."

What is the main flaw?

A. It counts too few books.

B. It reports a positive association.

C. It treats an association as a cause.

D. It uses a line of fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

True or false?

Predictions far outside the range of the data are less reliable than predictions inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

True or false?

A survey of volunteers always represents the whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 8.SP.A.4 • STATISTICS & PROBABILITY

Construct a two-way table summarizing data on two categorical variables

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Report the number of observations in a data set	**6.SP.B.5.a**
BACKWARD 2	Solve "how many more" and "how many in all" problems with a sc…	**3.MD.B.3**
FORWARD 1	Find joint, marginal, and conditional relative frequencies in a two…	**HSS-ID.B.5**
FORWARD 2	Use a two-way table to find conditional probabilities and decide…	**HSS-CP.A.4**

SET 57 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

60 students were asked whether they have a curfew and whether they have chores. 36 have a curfew. 24 of the students with a curfew have chores. 8 of the students without a curfew have chores.

Complete the two-way table.

	Chores	No chores	Total
Curfew			
No curfew			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

50 students were asked whether they prefer summer or winter. 22 are in Grade 7 and 28 are in Grade 8. 15 of the Grade 7 students and 18 of the Grade 8 students prefer summer.

Complete the two-way table.

	Summer	Winter	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

90 people were asked whether they exercise daily and whether they eat breakfast. 40 exercise daily. 30 of the people who exercise daily eat breakfast. 50 of the 90 people eat breakfast.

Complete the two-way table.

	Breakfast	No breakfast	Total
Exercises			
Does not			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

Ten students answered two questions: "Do you have a pet?" and "Do you have a sibling?" Their answers (pet, sibling) are:

yes–yes, yes–no, no–no, yes–yes, no–yes, yes–yes, no–no, no–yes, yes–no, no–no

Complete the two-way table.

	Sibling	No sibling	Total
Pet			
No pet			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

True or false?

A completed two-way table for 60 students has exactly two rows, Curfew and No curfew, and every student is in one of them. Its row totals can be 36 and 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 1

Report the number of observations in a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In a survey, 12 students chose soccer, 9 chose basketball, and 7 chose tennis.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

How many observations are shown in the dot plot?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

Each of 25 students named a favorite fruit. 5 different fruits were named.

How many observations are in the data set?

A. 5

B. 125

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

If 14 students answer yes and 11 answer no, there are 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

The data set 3, 3, 5, 7 has 3 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

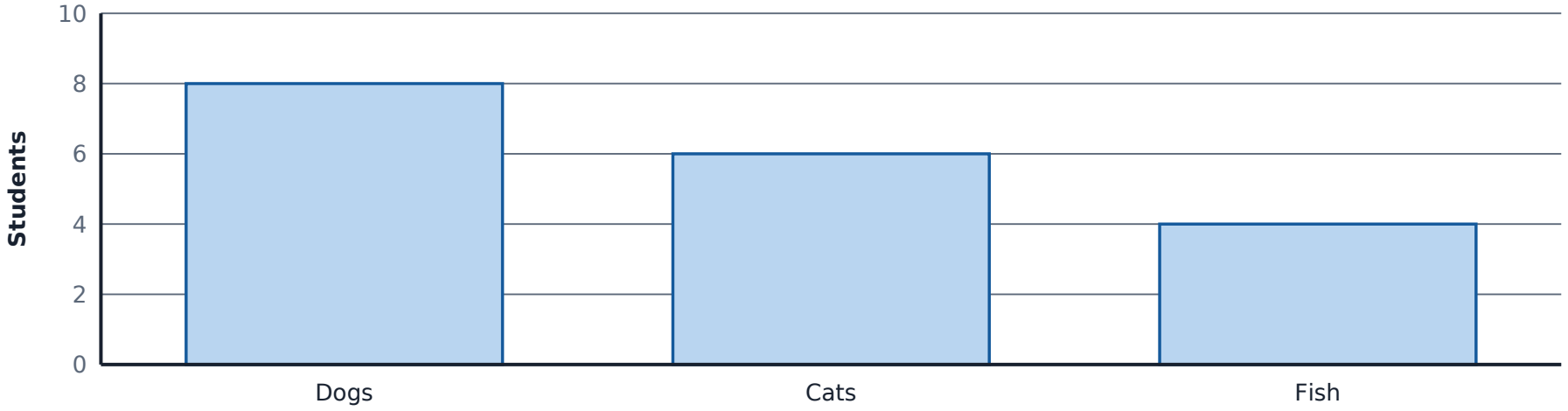
Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 2

Solve "how many more" and "how many in all" problems with a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many more students chose dogs than fish?



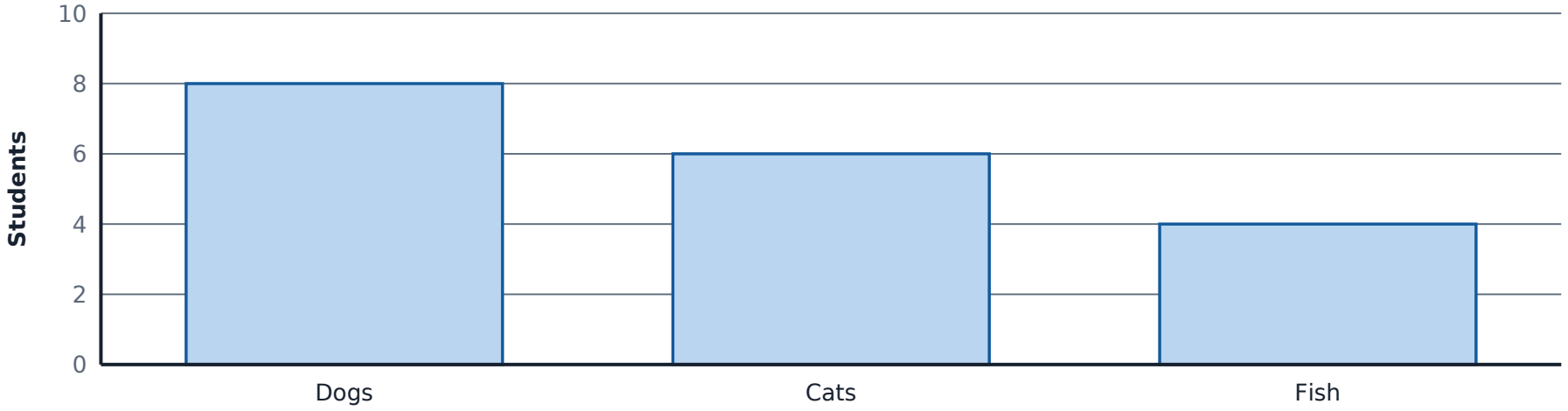
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

How many students chose a pet in all?



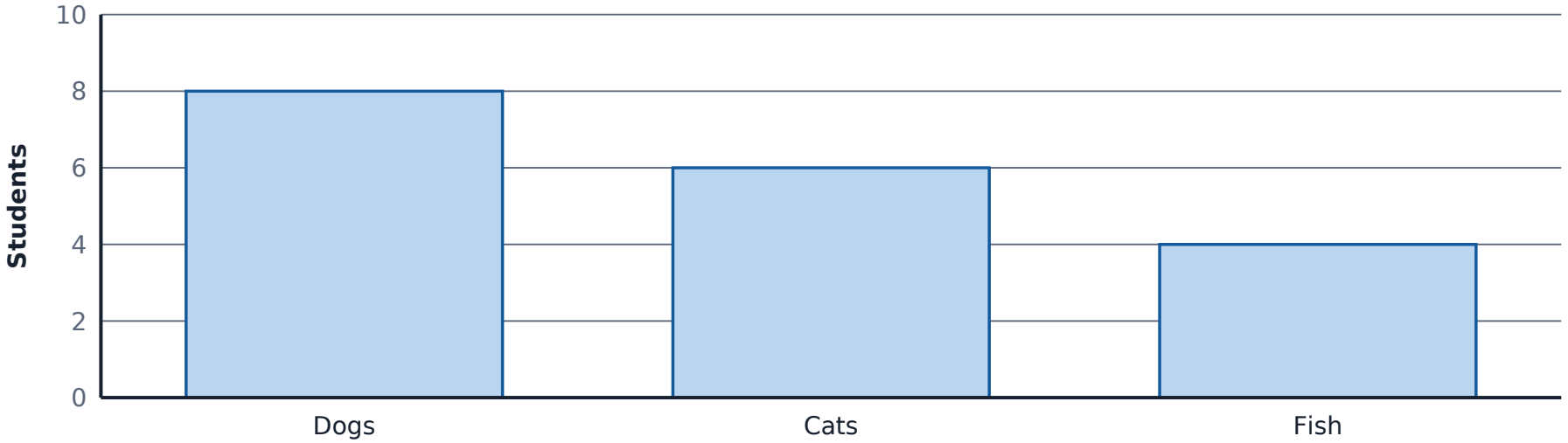
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

How many fewer students chose fish than cats?



A. 4 **B.** 6 **C.** 10 **D.** 2

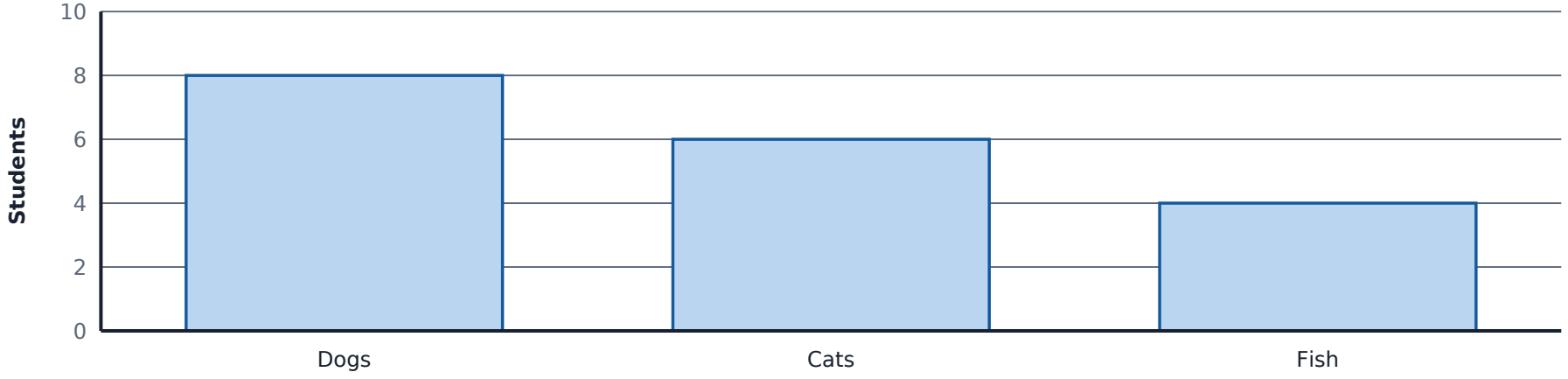
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

True or false?

Fewer students chose cats than fish.



A. True

B. False

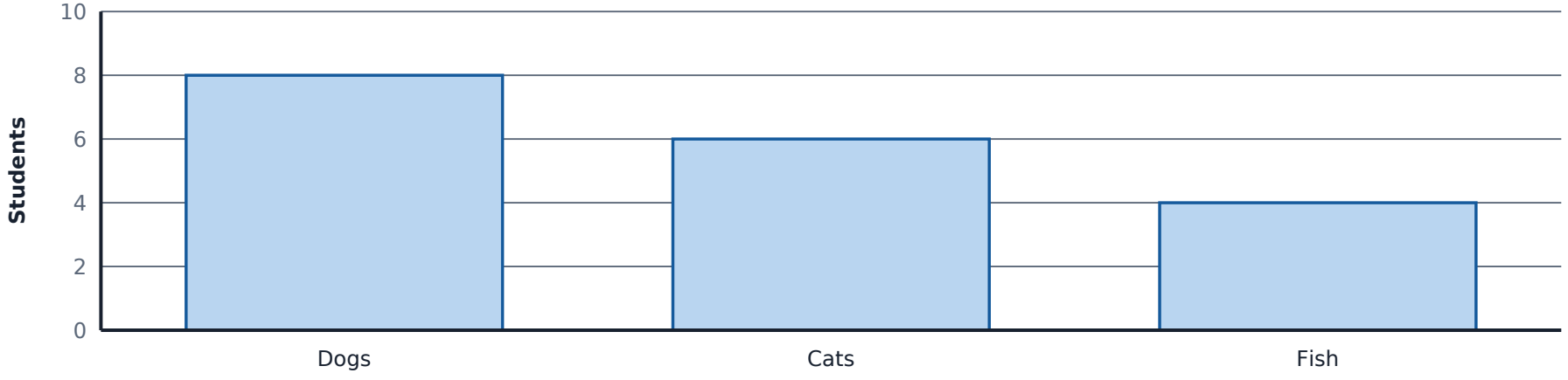
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

True or false?

The same number of students chose dogs and cats.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 1

Find joint, marginal, and conditional relative frequencies in a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

What is the joint relative frequency of students who have a curfew and have chores?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

What is the marginal relative frequency of students who have chores? Round to the nearest hundredth.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

What is the conditional relative frequency of having chores among students with no curfew?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. $\frac{2}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

True or false?

The marginal relative frequency of students with a curfew is 0.36.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

True or false?

The joint relative frequencies in the four inner cells add to 2.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 2

Use a two-way table to find conditional probabilities and decide whether events are independent • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student from the table is chosen at random.

Find P(in club | Grade 8).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

Are being in the club and being in Grade 7 independent? Explain with conditional probabilities.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

A student from the table is chosen at random.

Find P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. 0.6 **B.** 0.12 **C.** 0.4 **D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

True or false?

For a student chosen at random from the table, P(not in club) = 0.7.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

True or false?

For the table, P(in club | Grade 7) equals P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 8.SP.A.4 • STATISTICS & PROBABILITY

Use relative frequencies in a two-way table to describe a possible association

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Find a percent of a quantity and the percent one number is of ano…	**6.RP.A.3.c**
BACKWARD 2	Use ratio language to describe two quantities	**6.RP.A.1**
FORWARD 1	Recognize conditional probability and independence in everyday…	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as the fraction of B's outcomes that…	**HSS-CP.B.6**

SET 58 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

For each row, what percent of the people use a phone at night?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

Is there evidence of an association between sleep and phone use at night? Explain using the row relative frequencies.

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

Which conclusion do the row relative frequencies support?

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. There is a strong association because 45 is greater than 30.

B. Grade 7 students are more likely to eat breakfast.

C. Grade 8 students are more likely to eat breakfast.

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

What percent of the people who use a phone at night sleep under 8 hours?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

True or false?

Grade 8 students are more likely to eat breakfast than Grade 7 students, because 45 is greater than 30.

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 1

Find a percent of a quantity and the percent one number is of another • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What percent of 40 is 12?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

Find 60% of 75.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

36 out of 48 is what percent?

A. 75%

B. 48%

C. 12%

D. 36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

True or false?

15 out of 60 is 15%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

True or false?

20% of 50 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 2

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

In a group, 12 people use a phone at night and 28 do not.

Write the ratio of users to non-users in simplest form.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

A class has 18 students who walk to school and 12 who ride.

For every 3 students who walk, how many ride?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

In a survey, 30 people eat breakfast and 20 do not.

Which statement describes the ratio?

A. For every 3 who eat breakfast, 2 do not.

B. For every 1 who eats breakfast, 30 do not.

C. For every 3 who eat breakfast, 5 do not.

D. For every 2 who eat breakfast, 3 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

True or false?

The ratio 24 : 16 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The ratio 10 : 15 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 1

Recognize conditional probability and independence in everyday situations • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain the difference between P(sleeps under 8 hours | uses a phone at night) and P(uses a phone at night | sleeps under 8 hours).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

A fair coin is flipped twice, and the flips are independent.

Is P(second flip is heads | first flip is heads) equal to P(second flip is heads)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

Which pair of events is most likely independent?

A. Smoking and getting lung disease

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

C. Rain today and carrying an umbrella today

D. Studying more and getting a higher grade

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

If two events are independent, knowing that one happened does not change the probability of the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 2

Find a conditional probability as the fraction of B's outcomes that are also in A • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is even, what is the probability that it is greater than 3?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

Two fair coins are flipped, and the flips are independent.

Given that at least one coin shows heads, what is the probability that both show heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

Of 100 people, 48 use a phone at night, and 36 of those sleep under 8 hours.

A night phone user is chosen at random. What is P(sleeps under 8 hours | uses a phone at night)?

A. 0.75

B. 0.36

C. 0.48

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10.

Given that the number is odd, the probability that it is 5 or less is $\frac{3}{5}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

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True or false?

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is less than 3, the probability that it is 1 is $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5